

3.12.2 Wave-particle duality (A-level only)

3.12.2.1 Newton's corpuscular theory of light (A-level only)

Content

Comparison with Huygens' wave theory in general terms.

The reasons why Newton's theory was preferred.

3.12.2.2 Significance of Young's double slits experiment (A-level only)

Content

Explanation for fringes in general terms, no calculations are expected.

Delayed acceptance of Huygens' wave theory of light.

Nature of electromagnetic waves.

Maxwell's formula for the speed of electromagnetic waves in a vacuum $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

where μ_0 is the permeability of free space and ϵ_0 is the permittivity of free space.

Students should appreciate that ϵ_0 relates to the electric field strength due to a charged object in free space and μ_0 relates to the magnetic flux density due to a current-carrying wire in free space.

Hertz's discovery of radio waves including measurements of the speed of radio waves.

Fizeau's determination of the speed of light and its implications.

3.12.2.4 The discovery of photoelectricity (A-level only)

Content

The ultraviolet catastrophe and black-body radiation.

Planck's interpretation in terms of quanta.

The failure of classical wave theory to explain observations on photoelectricity.

Einstein's explanation of photoelectricity and its significance in terms of the nature of electromagnetic radiation.

3.12.2.5 Wave-particle duality (A-level only)

Content

de Broglie's hypothesis: $p = \frac{h}{\lambda}$;

$$\lambda = \frac{h}{\sqrt{2meV}}$$

Low-energy electron diffraction experiments; qualitative explanation of the effect of a change of electron speed on the diffraction pattern.

3.12.2.6 Electron microscopes (A-level only)

Content

Estimate of anode voltage needed to produce wavelengths of the order of the size of the atom.

Principle of operation of the transmission electron microscope (TEM).

Principle of operation of the scanning tunnelling microscope (STM).